AIRCRAFT PERFORMANCE – EXAMPLE PROBLEMS

Complete Solutions

PROBLEM 1: Glide Performance of Two Sailplanes

Theoretical Calculation:

Given:

- Wing reference area, $S = 15\ \mathrm{m}^2$
- Drag polar: $C_D = 0.012 + 0.030\,C_L^2 \quad \Rightarrow \quad C_{D0} = 0.012,\ K = 0.030$
- Release height, $h = 2000\ \mathrm{m}$ (calm air)
- Glider A: $W_A = 3750\ \mathrm{N}$, $W_A/S = 250\ \mathrm{N/m^2}$
- Glider B: $W_B = 3750 + 1500 = 5250\ \mathrm{N}$, $W_B/S = 350\ \mathrm{N/m^2}$
- Both fly at the best glide angle (maximum L/D)
- Air density (not given, standard sea-level value assumed): $\rho = 1.225\ \mathrm{kg/m^3}$

To find: (1) $R_A,\ R_B$ (2) $t_A,\ t_B$

Step 1: Lift coefficient for best glide

$$C_L^* = \sqrt{\frac{C_{D0}}{K}}$$

where, C_L^* = lift coefficient at $(L/D)_{max}$, C_{D0} = zero-lift drag coefficient, K = induced-drag factor

$$C_L^* = \sqrt{\frac{0.012}{0.030}} = \sqrt{0.4}$$
$$= 0.6325$$

Step 2: Drag coefficient at best glide

$$C_D^* = C_{D0} + K\,(C_L^*)^2 = 2\,C_{D0}$$
$$C_D^* = 0.012 + 0.030 \times (0.6325)^2 = 0.012 + 0.012$$
$$= 0.024$$

Step 3: Maximum lift-to-drag ratio

$$\left(\frac{L}{D}\right)_{max} = \frac{C_L^*}{C_D^*} = \frac{1}{2\sqrt{C_{D0}\,K}}$$

where, L = lift, D = drag

$$\left(\frac{L}{D}\right)_{max} = \frac{0.6325}{0.024} = 26.35$$

$$\text{Check: } \frac{1}{2\sqrt{0.012 \times 0.030}} = \frac{1}{2 \times 0.01897} = 26.35\ \checkmark$$

Step 4: Best glide angle

$$\tan\gamma_{min} = \frac{1}{(L/D)_{max}}$$

where, γ = glide (flight-path) angle below the horizontal

$$\tan\gamma_{min} = \frac{1}{26.35} = 0.03795$$
$$\gamma_{min} = \tan^{-1}(0.03795) = 2.173^\circ$$
$$\sin\gamma = 0.03792, \qquad \cos\gamma = 0.99928$$

Step 5: Maximum glide (ground) distance — Part 1

$$R = \frac{h}{\tan\gamma} = h\,(L/D)_{max}$$

where, R = horizontal distance covered, h = release height

$$R = 2000 \times 26.35$$
$$R_A = R_B = 52{,}705 \text{ m} \approx 52.70 \text{ km}$$

> **Result:** $R_A = R_B \approx 52.7$ km. Best-glide distance depends only on $(L/D)_{max}$, not on weight, so the water ballast does not change the distance.

Step 6: Best-glide airspeed

$$V = \sqrt{\frac{2\,(W/S)}{\rho\, C_L^*}}$$

where, V = airspeed along the flight path, W/S = wing loading, ρ = air density

Glider A ($W/S = 250$ N/m^2):

$$V_A = \sqrt{\frac{2 \times 250}{1.225 \times 0.6325}} = \sqrt{\frac{500}{0.7748}} = \sqrt{645.36} = 25.40 \text{ m/s}$$

Glider B ($W/S = 350$ N/m^2):

$$V_B = \sqrt{\frac{2 \times 350}{1.225 \times 0.6325}} = \sqrt{\frac{700}{0.7748}} = \sqrt{903.51} = 30.06 \text{ m/s}$$

Step 7: Sink rate and time airborne — Part 2

$$V_v = V\sin\gamma, \qquad t = \frac{h}{V_v} = \frac{h}{V\sin\gamma}$$

where, V_v = rate of descent (sink speed), t = time to touchdown

Glider A:

$$V_{v,A} = 25.40 \times 0.03792 = 0.9633 \text{ m/s}$$
$$t_A = \frac{2000}{0.9633} = 2076.2 \text{ s} \approx 34.60 \text{ min}$$

Glider B:

$$V_{v,B} = 30.06 \times 0.03792 = 1.1398 \text{ m/s}$$

$$t_B = \frac{2000}{1.1398} = 1754.7 \text{ s} \approx 29.24 \text{ min}$$

Final answers — Problem 1

1) $R_A = R_B = 52.70$ km

2) $t_A = 2076$ s (34.6 min), $t_B = 1755$ s (29.2 min)

Glider B flies faster (30.1 vs 25.4 m/s), so it lands sooner but covers the same distance.

Check: $t_A / t_B = \sqrt{350/250} = 1.183$.

PROBLEM 2: Cruise Range and Endurance of a Business Jet

Theoretical Calculation:

Given:

- $W_0 = 100{,}000$ N (start of cruise), $W_1 = 80{,}000$ N (empty fuel)
- Wing area, $S = 30\ \text{m}^2$
- Drag polar: $C_D = 0.016 + 0.045\,C_L^2 \quad \Rightarrow \quad C_{D0} = 0.016,\ K = 0.045$
- $c_t = 0.65\ \text{hr}^{-1} = 1.805 \times 10^{-4}\ \text{s}^{-1}$ (constant)
- Flight 1: sea level, $\rho_{SL} = 1.225\ \text{kg/m}^3$
- Flight 2: 33,000 ft, $\rho_{33k} = 0.380\ \text{kg/m}^3$
- Both flown at maximum-range condition (jet, constant c_t)

To find: (1) $R_{SL},\ R_{33k}$ in km (2) $E_{SL},\ E_{33k}$

Step 1: Lift coefficient for maximum range (jet)

For a jet, range is maximum when $\sqrt{C_L}/C_D$ is maximum, which gives:

$$C_{L,mr} = \sqrt{\frac{C_{D0}}{3K}}$$

where, $C_{L,mr}$ = lift coefficient for maximum range

$$C_{L,mr} = \sqrt{\frac{0.016}{3 \times 0.045}} = \sqrt{\frac{0.016}{0.135}} = \sqrt{0.11852}$$
$$= 0.3443$$

Step 2: Drag coefficient and L/D at this condition

$$C_D = C_{D0} + K\,C_L^2 = C_{D0} + K \cdot \frac{C_{D0}}{3K} = \frac{4}{3}\,C_{D0}$$

$$C_D = 0.016 + 0.045 \times (0.3443)^2 = 0.016 + 0.005333 = 0.021333$$

$$\frac{L}{D} = \frac{C_L}{C_D} = \frac{0.3443}{0.021333} = 16.137$$

$$\frac{\sqrt{C_L}}{C_D} = \frac{\sqrt{0.3443}}{0.021333} = \frac{0.5867}{0.021333} = 27.503$$

Step 3: Range equation (Breguet, constant c_t and C_L)

$$R = \frac{2}{c_t}\sqrt{\frac{2}{\rho S}}\,\frac{\sqrt{C_L}}{C_D}\left(\sqrt{W_0} - \sqrt{W_1}\right)$$

where, R = range, c_t = thrust specific fuel consumption (s^{-1}), ρ = air density, S = wing area, W_0, W_1 = initial and final weights

Common terms:

$$\sqrt{W_0} - \sqrt{W_1} = \sqrt{100000} - \sqrt{80000} = 316.228 - 282.843 = 33.385$$

$$\frac{2}{c_t} = \frac{2}{1.805 \times 10^{-4}} = 11{,}080 \text{ s}$$

Flight 1 — Sea level ($\rho = 1.225 \text{ kg/m}^3$):

$$\sqrt{\frac{2}{\rho S}} = \sqrt{\frac{2}{1.225 \times 30}} = \sqrt{0.05442} = 0.23328$$

$$\begin{aligned} R_{SL} &= 11{,}080 \times 0.23328 \times 27.503 \times 33.385 \\ &= 2{,}373{,}444 \text{ m} \\ R_{SL} &= 2{,}373.4 \text{ km} \end{aligned}$$

Flight 2 — 33,000 ft ($\rho = 0.38 \text{ kg/m}^3$):

$$\sqrt{\frac{2}{\rho S}} = \sqrt{\frac{2}{0.38 \times 30}} = \sqrt{0.17544} = 0.41885$$

$$\begin{aligned} R_{33k} &= 11{,}080 \times 0.41885 \times 27.503 \times 33.385 \\ &= 4{,}261{,}429 \text{ m} \\ R_{33k} &= 4{,}261.4 \text{ km} \end{aligned}$$

> **Result:** $R_{SL} \approx 2{,}373$ km, $R_{33k} \approx 4{,}261$ km. Check: $R_{33k}/R_{SL} = \sqrt{1.225/0.380} = 1.795$ ✓ (same L/D, but at higher altitude the jet flies faster for the same C_L).

Step 4: Endurance while flying the max-range profile

Fuel burn: $dW/dt = -c_t\, D = -c_t\, W/(L/D)$. With constant L/D (constant C_L), integrating gives:

$$E = \frac{1}{c_t}\left(\frac{L}{D}\right)\ln\left(\frac{W_0}{W_1}\right)$$

where, E = endurance (time airborne), L/D = lift-to-drag ratio at the flown condition

$$\ln\left(\frac{W_0}{W_1}\right) = \ln\left(\frac{100000}{80000}\right) = \ln(1.25) = 0.22314$$

$$\begin{aligned} E &= \frac{1}{1.805 \times 10^{-4}} \times 16.137 \times 0.22314 \\ &= 5{,}540.2 \times 16.137 \times 0.22314 \\ E &= 19{,}950 \text{ s} = 5.54 \text{ hr} \end{aligned}$$

ρ does not appear in this formula, so endurance is the same at both altitudes:

$$E_{SL} = E_{33k} = 19{,}950 \text{ s} \approx 5.54 \text{ hr}$$

Final answers — Problem 2

1) $R_{SL} = 2{,}373.4$ km, $R_{33k} = 4{,}261.4$ km

2) $E_{SL} = E_{33k} = 5.54$ hr ($19{,}950$ s)

Altitude increases range (higher speed at same C_L) but leaves endurance unchanged.